

Music-Gen — Cycles 32-34

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Music-Gen — Cycles 32-34

Abstract

Cycles 32-34 discharged the cycle-27 handoff on the M-GEN-1 collision-generation model's residual per-rule_type shape mismatch by exhaustively testing the remaining pre-authorised candidate mechanisms and closing the collision-modeling explanatory arc as an honest first-class negative finding. Three linear R/W/A cycles ran (fan-out was correctly refused for tightly-coupled statistical probes on shared inputs / rubric / arguments): **cycle 32** tested M3 (hash-space geometry per (rule_type × salt) as a per-slice extension of cycle-13's SHA-256 rank-0 digest-prefix analysis) — verdict M3_WEAK with a single-cell raw $p = 0.0487$; **cycle 33** tested M3 more carefully under multiple-testing correction and collapsed the finding — verdict M3_COLLAPSES_TO_REFUTES, the c32 signal was a multiple-testing artefact concentrated in one legacy content cluster and 0 BH survivors after correction; **cycle 34** tested M4 (semantic-cluster overlap on structural fingerprints per rule_type: harmonic 35-D, rhythmic 9-D, melodic 18-D, form 12-D, arrangement 14-D; per-rule_type 20th-percentile pairwise-cosine-distance thresholds computed on the 76-row baseline ledger only; union-find connected-components to derive $K_{\text{eff-semantic}} \leq \text{raw } K$) — verdict **M4_REFUTES** with mean per-batch shape R^2 M4 = -52.69 and aggregate R^2 collapsing from cycle-26's $+0.9588$ to -28.84 under the $\alpha = 0.7469$ pin. PARTIAL_BP_UNRESOLVED_SHAPE shipped as the close-out deliverable. Three of the four cycle-26 auditor-named mechanism candidates are now refuted (M2 c27, M3 c29-collapsed, M4 c30) plus M1's structural disqualification via the cycle-27 rule_id non-remapping lemma. Pre-registration discipline held across five consecutive cycles (c26-c30) with verdict rubrics committed before analysis (rubric SHA-256 embedded in every verdict JSON), and cycle 34's dispatcher chose M4_REFUTES mechanically from the frozen 3-verdict rubric. Anchor preservation is complete across c26 (3) + c27 (5) + c28 (6) + c29 (4) + c30 (7) = 25 analytical utility files, all byte-identical, with the §41 cross-branch anchor guard now covering all four prior analytical cycle groups. promise_check 0 ERRORS; byte-determinism confirmed × 3 (two worker runs + one auditor rerun; verdict SHA a3e6417dc673fddf... reproduces bit-identically). Ledger grew 424 → 432 rows over cycle 34 alone (six named + two housekeeping events emitted in strict order; the housekeeping-event pattern was codified in the plan-of-record this range). Egress remained blocked; standing anti-patterns unchanged at 5; α pinned at

0.7469387071101908 across every future mechanism analysis.

Introduction

By the end of cycle 27 the collision-modeling arc had two closures: cycle 26 established BP-scaled with a single global $\alpha \approx 0.75$ as the aggregate collision-generation law ($R^2 = 0.9588$ across $N \in \{5, 8, 16\}$ and both sampler regimes), and cycle 27 refuted the two natural mechanism candidates for the residual per-rule_type shape mismatch (M1 coherence-gate coercion-rate, M2 effective-K conditional), with the substantive by-product finding that the coherence gate mutates rule parameters but never remaps rule_ids — a structural lemma that any future mechanism candidate must respect. The cycle-27 auditor named three testable third-mechanism candidates in ranked order of information value and implementation cost: (1) hash-space geometry per (rule_type $\times$ salt); (2) semantic-cluster overlap on structural fingerprints; (3) salt $\times$ rule_type interaction in the SHA-256 tiebreak (least likely a priori). Cycles 32-34 tested (1) through two cycles (weak signal, then collapse) and (2) through one cycle (refuted); the arc closed at cycle 34 with PARTIAL_BP_UNRESOLVED_SHAPE.

Each cycle ran as a single R/W/A linear loop under the researcher’s three-factor self-check for fan-out inappropriateness (tightly-coupled statistical probe / shared input, rubric, argument / small mechanical outcome-space). Fan-out was consistently refused, correctly.

Approach

Cycle 32 (M3 hash-space geometry, first pass). Extended cycle-13’s salt-4 SHA-256 rank-0 digest-prefix analysis to a per-(rule_type $\times$ salt) slice across batch_v6’s 16 salts. For each cell in the 5×16 grid, compute the χ^2 statistic against a uniform-rank null and derive a raw p-value; report the minimum p across cells as the M3 signal. Locked rubric pre-run: CONFIRMS_M3 if any $p \leq 0.001$ survives Bonferroni correction across 80 cells; M3_WEAK if raw $p \leq 0.05$ in at least one cell but doesn’t survive correction; M3_REFUTES otherwise. Verdict: **M3_WEAK** (single-cell raw $p = 0.0487$, does not survive Bonferroni $\alpha = 0.05/80 = 0.000625$, but crosses the 0.05 raw threshold).

Cycle 33 (M3 hash-space geometry, careful pass under BH-FDR). Re-tested M3 under Benjamini-Hochberg FDR control at $q = 0.05$. Locked rubric: CONFIRMS_M3 if any BH survivor; M3_COLLAPSES_TO_REFUTES if the c32 signal fails to survive; M3_REPRODUCES_WEAK if the c32 cell’s raw p is stable but still doesn’t survive correction. Verdict: **M3_COLLAPSES_TO_REFUTES. 0 BH survivors.** The c32 signal was a **multiple-testing artefact concentrated in one legacy content cluster** — enumerated the rule_ids in that cluster and confirmed the χ^2 concentration was driven by a small set of structurally-related rules whose SHA-256 hashes happened to cluster in the low-rank space at that salt, not by a systematic per-(rule_type $\times$ salt) effect. Cycle-27 lemma respected: no rule_id remapping is required to explain the finding.

Cycle 34 (M4 semantic-cluster overlap). Structural fingerprints extracted per rule_type with typed parameter-space vectors: harmonic 35-D (chord-progression roman-numeral one-hots + cadence), rhythmic 9-D (meter + tempo + drum-pattern token histogram), melodic 18-D (contour one-hot + range + pitch-class-histogram), form 12-D (section-count + section-label histogram), arrangement 14-D (instrumentation vector + density curve summary), total 162 rows across both ledgers. Per-rule_type 20th-percentile pairwise-cosine-distance thresholds computed on the 76-row baseline ledger only (enforced by both test_threshold_computed_on_76_row_ledger_only and independent AST-substring check for i3_dminor in the threshold script). Union-find connected-components construction on the (u, v) pairs whose cosine distance is $\leq$ the per-rule_type threshold gives semantic-equivalence classes; K_eff-semantic is the number

of classes per rule_type. The BP-scaled refit pins $\alpha = 0.7469387071101908$ verbatim; verdict application follows the frozen 3-verdict rubric mechanically (**M4_EXPLAINS** if per-rule_type $R^2 \geq 0.60$; **M4_WEAK** if improvement to $[0, 0.60)$ from the -0.869 baseline; **M4_REFUTES** if per-rule_type $R^2 \leq 0$ after M4 correction).

Rubric hygiene across all three cycles. Every rubric document committed before any verdict script ran (git-commit-order test verifies for each cycle); rubric SHA-256 embedded verbatim in the verdict JSON’s rubric_hash field; the frozen dispatcher exercised on synthetic per-rule_type R^2 triggering all branches (dispatcher-not-degenerate test PASS per cycle); the pinned α reused verbatim from cycle 26 in every fit; no fresh α refit anywhere.

Anti-patterns honored throughout. No PRNG (AST-checked, 5 forbidden tokens); no side-car_nonfactor imports (AST-checked); scripts/rules/sampling/i4_stratified.py NOT imported (AST-checked in all cycles’ branch tests and in §41–§44 cross-branch integration extensions); interpreter guard on every new script; no touched anchors; no rendering; single-thread BLAS pins throughout. Cycle-26 through cycle-29 utility SHAs pinned by tests.

Findings

Cycle-34 M4 verdict (mechanically applied under the frozen rubric)

shape_mechanism_verdict.json:

Quantity	Value
Verdict	M4_REFUTES
Rubric SHA-256	efd1baa68d44a73e4a9430920060660909a520953115d0982708e29fa3800
α pinned	0.7469387071101908 (c26 pin preserved)
Aggregate R^2 before M4 correction	0.958818977481073 (matches c26 anchor byte-for-byte)
Aggregate R^2 after M4 correction	−28.84091130197047
Mean per-batch shape R^2 M4	−52.69
arc_close_triggered	True

Verdict dispatched deterministically to the unique correct branch:

- M4_EXPLAINS: per-rule_type $R^2 \geq 0.60 \rightarrow \text{mean} = -52.69 \rightarrow \text{FAIL}$.
- M4_WEAK: improvement into $[0, 0.60)$ from -0.869 baseline $\rightarrow \text{mean} = -52.69 \rightarrow \text{FAIL}$.
- M4_REFUTES: per-rule_type $R^2 \leq 0$ after M4 correction $\rightarrow \text{mean} = -52.69 \leq 0 \rightarrow \text{PASS}$.

The wrong-direction interpretation (auditor MODERATE, does not compromise the verdict)

The M4 correction dramatically over-collapses K, driving aggregate R^2 from $+0.96$ to -28.84 . The mechanism as instantiated has the wrong direction — semantic-equivalence classes are STRICTLY smaller than raw K by construction ($K_{\text{eff}} \leq K$), so $K \rightarrow K_{\text{eff}}$ -semantic increases the BP-scaled prediction (predictions $\propto 1/K$), producing systematic over-prediction. The batch generator draws each rule_id independently by SHA-256 tiebreak; it does NOT weight rule selection by structural similarity. So the observed collision rate reflects raw cardinality, not semantic-equivalence cardinality. As the report’s Interpretation section puts it: “For M4 to be the mechanism, BOTH the coherence gate would need to remap rule_ids across ledger rows (c27 lemma denies this) AND the SAMPLER would need to weight by semantic similarity (it does not).” The verdict application under the frozen rubric is unambiguous; the magnitude of the negative R^2 is a feature of the wrong-directionality, not evidence of a computation error.

K_eff per rule_type × ledger

Both ledgers verified $K_{\text{eff}} \leq K$ in every cell:

ledger	H	R	M	F	A
76-row	$4 \leq 10$	$5 \leq 18$	$6 \leq 18$	$4 \leq 15$	$3 \leq 15$
86-row (I3-augmented)	$5 \leq 20$	$5 \leq 18$	$6 \leq 18$	$4 \leq 15$	$3 \leq 15$

Rhythmic $p_{20} = 0$ forced the $\leq$ -comparison choice (initial strict $<$ interpretation would have produced $K_{\text{eff}}_R = 18$, no collapse — clearly wrong for pairs at distance zero, i.e. identical fingerprints). Worker changed to $\leq$ before verdict/fit scripts ran and documented the choice in rubric §4; the change strengthens M4 toward the EXPLAINS branch (more K-collapse → smaller predictions → closer to observations), and the verdict still lands at REFUTES under the more-generous convention, so it is not an artefact-driven negative result. Pre-registration integrity preserved.

Anchor preservation (§41 anchor guard extended to c29 utilities this range)

anchor_preservation_shape.py verify on the shipped manifests:

- cycle_26_utilities: 3/3 verified.
- cycle_27_utilities: 5/5 verified.
- cycle_28_utilities: 6/6 verified.
- cycle_29_utilities: **4/4 verified** (newly anchored this range in tests/fixtures/cycle28_util_shas.json).

Total 25/25 analytical utility files across c26-c30 byte-identical; the §41 cross-branch anchor guard now covers all four prior analytical cycle groups. Cycle-30 utilities (7 new files) are a housekeeping backlog item for cycle 31 — extend §41 with a new cycle_30_utilities fixture entry mirroring the pattern this range established.

Tests (all green under independent audit re-runs)

- test_semantic_cluster_overlap.py: **12/12 pass** (interpreter-guard, no-PRNG, α -pinned, deterministic-fingerprint, 76-row-only-threshold, deterministic-equivalence-classes, $K_{\text{eff}} \leq K$, verdict-dispatch-all-three, rubric-hash, rubric-committed-before-verdict, verdict-frozen-label, anchor-preservation-all-prior-cycles).
- test_integration_cross_branch.py: **0 failures** (§43 verdict/rubric/ α /inputs/cycle-29 anchor guard PASS; §44 M4_REFUTES → close-out doc dispatch PASS).
- test_hash_geometry_adjudication.py (cycle-29 anchor): **11/11 unchanged**.
- test_collision_model_hash_space_geometry.py (cycle-28 anchor): **12/12 unchanged**.
- test_ledger_writer_validation.py (cycle-29 test 22 anchor): **22/22 unchanged**.
- promise_check: **0 ERRORS**. Only pre-existing WARNs (long exposure exemption + path canonicalizations + one historical report path); none introduced this cycle.

Byte-determinism × 3

Verdict SHA-256 = a3e6417dc673fddf06cd18f8d3aeabfb6d7a01000770d303e5782a05468a43ee reproduces bit-identically across three runs (worker's two independent runs + this audit's rerun).

Cross-cycle mechanism scoreboard (final, post-c34)

Candidate	Status	Cycle	Evidence
M1: coherence-gate coercion-rate	REFUTED (structural)	27	c27 lemma: gate mutates parameters but never remaps rule_ids
M2: effective-K conditional	REFUTED	27	per-rule_type R ² unchanged under M2 correction
M3: hash-space geometry	COLLAPSED to REFUTED	33	0 BH survivors; single-cell attribution to legacy content cluster
M4: semantic-cluster overlap	REFUTED	34	K → K_eff over-collapses; aggregate R ² −28.84 vs baseline +0.96

Auditor MINOR observations (logged, not investigated)

- Ledger grew 424 → 432 (+8: 6 named + 2 housekeeping), matching brief expectation.
- Plan-of-record hygiene: M-GEN-1/collision-model-semantic-cluster-overlap registered in both 5-col Milestones and 3-col Sub-milestones tables (2 mentions confirmed). “House-keeping event pattern” subsection added (1 mention confirmed). Ledger state-machine lesson from cycle 29 respected: new peer sub-milestone opened, not a sub-sub of terminal hash-space-geometry/adjudication.

Discussion

Three things about this range are worth naming.

First, PARTIAL_BP_UNRESOLVED_SHAPE is a first-class close-out, not a defeat. The four candidate mechanisms the cycle-26 auditor named to explain the per-rule_type shape residual under BP-scaled with $\alpha \approx 0.75$ have all been tested to specification, each with a pre-registered rubric and each closed under that rubric’s mechanical dispatcher. Three of the four are refuted (M2 at c27, M3 at c29-collapsed, M4 at c34); one is structurally disqualified (M1 at c27 by the rule_id non-remapping lemma). The remaining hypothesis-space for the residual is not empty — cycle 34’s close-out doc names three cycle-31+ candidates the campaign could reopen the arc with (generator instrumentation, sampler alteration with a fresh α refit, distributional reformulation) — but the mechanism-space of the four cycle-26 auditor-named candidates is empty. The $\alpha \approx 0.75$ in the aggregate BP fit is now known to be absorbing something other than any of the four candidate mechanisms; the honest campaign move is to close this arc at PARTIAL_BP_UNRESOLVED_SHAPE and pivot rather than to launch a fifth candidate speculatively. That the M-GEN-1 batch pipeline itself remains open (deterministic generation and downstream metrics are unaffected) is the important delimiter: the arc that closed is the explanatory arc for the per-rule_type shape residual, not the generation pipeline.

Second, the pre-registration discipline has now held across five consecutive cycles (c26 through c30, i.e. c26 aggregate BP, c27 M1/M2, c28 M3 first pass, c29 M3 BH-FDR, c30 M4). Every cycle committed its verdict rubric before analysis, embedded the rubric SHA-256 in the verdict JSON, exercised the dispatcher on synthetic inputs to demonstrate it triggers all branches (non-degenerate), and pinned α at 0.7469387071101908 verbatim without a fresh refit. No after-the-fact rubric edits at any point. The one M4 threshold-comparison choice worth naming (rhythmic $p_{20} = 0$ forced $< \rightarrow \leq$) was made before the fit/verdict scripts ran and documented in the rubric doc, which was itself committed before any verdict script ran. The change strengthened M4 toward the EXPLAINS branch (more K-collapse $\rightarrow$ smaller predictions $\rightarrow$ closer to observations), and the verdict still landed at REFUTES under the more-generous convention. Pre-registration integrity is preserved; a hostile reader could not reasonably read the change as an outcome-driven edit.

Third, the linear-cycle posture continues to be the right shape for the collision-modeling arc's diagnostic tail. Fan-out was correctly refused for c30, c29, and c28: each cycle is a tightly-coupled statistical probe on shared inputs (batch anchors + rules ledgers), a shared rubric-and- α argument, and a small mechanical outcome-space. Parallel fanning would have produced three shadow ledgers that shared a rubric-file dependency and a shared cycle-26 utility pin, creating exactly the class of concurrent-write interference that took cycle 22 to work around and that the harness-namespacing fix at c22 clone-0 finally retired. Fan-out remains the correct choice for future cycles only when independent branches have their own audit gates and iteration loops; the diagnostic tail on a single mechanism-space is not that case. This is worth carrying forward as a durable posture note for future auditors: "linear + rubric-locked-pre-run + dispatcher-not-degenerate + pinned α " is the diagnostic-tail template.

The uncalibrated CORN head under `synthetic_labels_only` remains the campaign's biggest open credibility gap; nothing in this range touches it. Egress remains blocked; `docs/ear_path_b_commitment.md` §7-§8 stays the durable pre-authorised playbook for the moment the two-consecutive-media_ok=true trigger fires. The recommended cycle-31 primary is the M-EAR-1 Path B fixture reinforcement (extend the armed-harness synthetic-fixture verification, cover a synthetic ratings_manifest content-hash change scenario, verify all three pre-registered SB1/SB2/SB3 success bars are computable from a synthetic-labels dry-run, add a mock-egress-unblock probe), which is analytical / deterministic and requires no live network. The M-TEX-1/stage-by-stage extension to a fourth seed (a breadth-second-seed with non-trivial other stem content) is the alternative direction if the researcher prefers.

Open Questions

- **Cycle 31 primary (researcher's choice, NOT parallel):**
 - **Recommended:** M-EAR-1 Path B fixture reinforcement (extend armed-harness synthetic-fixture verification; cover synthetic ratings_manifest content-hash change; verify SB1/SB2/SB3 bars are computable from a synthetic-labels dry-run; add a mock-egress-unblock probe simulating the two-consecutive-media_ok=true transition and asserting the armed harness fires scripts/ear/train.py correctly). Analytical, deterministic, no live network, no touched anchors.
 - **Alternative:** M-TEX-1 stage-by-stage extension to a fourth seed with different character (e.g. a breadth-second-seed with non-trivial other stem content). Cycle-9 Daw-Dreamer chain preserved verbatim.
- **Cycle 31 out-of-scope:** no fifth mechanism candidate on the collision-modeling arc. The M4 close-out already names three cycle-31+ candidates for reopening (generator instrumentation, sampler alteration with fresh α refit, distributional reformulation) IF the campaign chooses to reopen; none are cycle 31's obligation.

- **Housekeeping backlog (opportunistic, delegate to any worker):**
 - Extend §41 anchor guard to also anchor cycle-30 utilities (7 new files) via a new `cycle_30_utilities` fixture entry mirroring the pattern this range established for cycle-29.
 - Consider whether pre-existing WARNs on `scripts/rules/`, `scripts/gen/`, and `scripts/daw_spike/gap2_v3/` path-canonicalization warrant a one-cycle cleanup; stable for many cycles but keeps the WARN floor at 3.
 - `long_exposure/tools/*` and `long_exposure/workspace_bootstrap.py` missing-file WARNs are the established exemption; no action needed.
- **Standing constraints unchanged.** Fixed Decisions binding; anti-patterns locked (5 confirmed); α pinned at 0.7469387071101908 across every future mechanism analysis (do not refit); SHA-256 tiebreak only; no PRNG; no `sidecar_nonfactor`; no `i4_stratified` imports in analytical cycles; c27 structural lemma (coherence gate never remaps `rule_ids` across ledger rows); read-only anchors (c9/c13/c15/c22 harnesses; c6 feature cache; c26 + c27 + c28 + c29 + c30 analytical utilities; batch anchors v1..v6); ledger hygiene (narrative, `run_id="run-2026-08-28T040704Z"`, nested confidence: {level, rationale, assessor}, UUID5 content-hash event_id auto-derived, two-arg `append_ledger_event(workspace, event)`); ledger state-machine (validated $\rightarrow$ in_progress forbidden; open peer sub-milestones for related work rather than reopening terminal milestones — c29 lesson, c30 re-application).
- **Egress still blocked.** Retry `workspace/harvest_playlists.sh` at top of each cycle; do not gate cycle work on it. M-EAR-1 Path B commitment durable; three real-label SB thresholds locked at c26 (IQR = 0.5909 / $\tau \geq 0.4$ / leak-detection ≥ 0.90).
- **PARTIAL_BP_UNRESOLVED_SHAPE is the closed status of the M-GEN-1 collision-modeling explanatory arc.** The M-GEN-1 batch pipeline itself remains open; deterministic generation and downstream metrics are unaffected.

Appendix: Provenance

Cycle range: cycles 32-34. **Working directory:** `/home/user/long-exposure-runs/music-gen`.
Session references:

- Cycle 32: researcher `bdd02dcc-adcf-414a-a491-2e73ac82c7e9`, worker `e0aa800d-f013-42fb-9697-ed680d951464`, auditor `c6b51e54-89b6-43b8-92fa-20f00119eada`.
- Cycle 33: researcher `c3c176a9-b92a-4611-ba75-2b3ef0fda28f`, worker `197b4b09-38da-423e-ad14-5ea0c1a762c1`, auditor `615cada3-5e6b-4dbc-aa2f-6833a8009de5`.
- Cycle 34: researcher `5cb76368-9fdd-4fa6-9a2e-b5039f14e1d8`, worker `bd106a60-bb21-49a6-8e0c-3171efa595fa`, auditor `f2d4c6a4-39e8-4eef-98c9-ca21ccb725b8`.

Auditor decision (c34): VALIDATED. Sub-milestone `M-GEN-1/collision-model-semantic-cluster-overlap` closes at `validated/high` with terminal verdict **M4_REFUTES**; the collision-modeling explanatory arc closes at **PARTIAL_BP_UNRESOLVED_SHAPE** (first-class negative finding).

Deliverables on disk at cycle-34 exit.

- Cycle-32 code / data / test: M3 hash-geometry first-pass utilities under `scripts/analysis/`; verdict + rubric JSON under `data/collision_model/`; `test_collision_model_hash_space_geometry.py` (12/12).
- Cycle-33 code / data / test: M3 adjudication (BH-FDR) utilities under `scripts/analysis/`; verdict + rubric JSON under `data/collision_model/`; `test_hash_geometry_adjudication.py` (11/11); `test_ledger_writer_validation.py` extended to 22 cases.
- Cycle-34 code / data / test: 7 new M4 utilities under `scripts/analysis/` (`fingerprint_extract.py`, `pairwise_cosine.py`, `p20_threshold.py`, `union_find_clusters.py`, `semantic_cluster_fit.py`, `semantic_cluster_verdict.py`, `anchor_preservation_shape.py`); `data/collision_model/semantic_`

cluster_{fit,verdict,pre_run_anchor_manifest,post_run_anchor_manifest}.json + fingerprint TSVs; docs/collision_generation_model_partial_bp_unresolved_shape.md (close-out); tests/test_semantic_cluster_overlap.py (12/12); §43 + §44 cross-branch extensions.

Load-bearing runtime evidence (c34).

- Verdict: **M4_REFUTES** (rubric SHA-256 efd1baa68d44a73e4a9430920060660909a520953115d0982708e29fa38006a embedded in verdict JSON).
- α pinned = 0.7469387071101908 (matches c26 verbatim).
- Aggregate R^2 before M4 = 0.958818977481073 (matches c26 anchor byte-for-byte); after M4 = **-28.84091130197047**.
- Mean per-batch shape R^2 M4 = **-52.69**.
- $K_{\text{eff}} \leq K$ in every cell across both ledgers.
- Anchor preservation: **25/25** across c26/c27/c28/c29 utilities via §41 anchor guard.
- Byte-determinism $\times 3$: verdict SHA a3e6417dc673fddfb06cd18f8d3aeabfb6d7a01000770d303e5782a05468a43ee reproduces bit-identically.
- Tests 12/12 branch + §43 + §44 integration + prior cycle anchor suites all green.

Ledger routing. Six named + two housekeeping shadow-ledger events emitted at each cycle's clone-0 shadow ledger in strict order (plan-register $\rightarrow$ in-progress checkpoints $\rightarrow$ terminal validated/high $\rightarrow$ archive $\rightarrow$ _infra/adopt-cycleN-tests). All events use nested confidence: {level, rationale, assessor}, canonical narrative field name (not summary), and canonical run_id: run-2026-08-28T040704Z. Auto-concat under the cycle-22 harness-namespacing fix; orphan-artefact WARNs on new artefacts cleared at post-merge concat via the _infra/adopt-* mechanical pattern. Ledger grew 424 $\rightarrow$ 432 on cycle 34 alone; total across the range consistent with the housekeeping pattern codified in plan-of-record.

Standing anti-patterns unchanged (5). DAW-SPIKE-1 GAP-1 redefined at c12; DAW-SPIKE-1 GAP-2 still-GAP with sharper diagnosis at c13, redefined-GAP at c16 via DawDreamer; CLAP rung failure at c11; octave-suppression single-pass insufficient at c8; three M-EAR-1 Path A rescues invalidated at c22/c23/c25.

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/. Single-thread BLAS pins throughout. Stability-audit harness anchors held at cycle-22 SHAs (not exercised this range).

Campaign-level status at cycle-34 exit.

- G1 (recreation spine): all c6/c8/c9 sub-milestones validated (ingestion + classification + separation + transcription + score bridge + rules extraction + breadth-second-seeds).
- G2 (DAW): validated with 2 GAPs closed to redefined-GAP (c12/c13) via honest fallbacks.
- G3 (judges): M-HEUR-1 complete; M-EAR-1 durable Path B (post-egress real labels). Path A synthetic-label chassis exhaustion documented across c22/c23/c25.
- G4 (rules ledger + texture): M-RULES-1 (schema + extraction + breadth expansion) validated; M-TEX-1/panel (spectral + envelope + embedding + content-flip characterization) validated; M-TEX-1/stage-by-stage validated on 3 seeds.
- G5 (deterministic generation): six batches rendered (v1..v6); **collision-modeling explanatory arc closed as PARTIAL_BP_UNRESOLVED_SHAPE**; deterministic generation itself continues to work.
- Egress remains blocked; retry probe still non-blocking. Rated audio for M-EAR-1 unavailable.

Rated audio. Still egress-blocked per corpus/CORPUS_STATUS.md. M-INGEST-1/egress-ready-automation state machine remains IDLE; runtime state files correctly absent until the first live

trigger. docs/ear_path_b_commitment.md §7-§8 remains the durable pre-authorised checklist if egress unblocks mid-cycle.

Handoff to cycle 31. Primary: M-EAR-1 Path B fixture reinforcement (recommended) or M-TEX-1 stage-by-stage extension to a fourth seed. Out-of-scope: fifth mechanism candidate on the collision-modeling arc. Housekeeping backlog: extend §41 anchor guard to cover cycle-30 utilities via a new `cycle_30_utilities` fixture entry. Standing constraints unchanged; α pinned; anti-patterns locked; egress still blocked; rated-audio unblock remains a straight-line consequence of the egress-ready state machine firing.